

# PUSHKARAJ BARADKAR

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## EDUCATION

**University of Southern California, Viterbi School of Engineering** GPA: 3.73 / 4.00

**Jan 2025 - Dec 2026**

**Master of Science in Computer Science**

Los Angeles, CA

Machine Learning for Data Science, Analysis of Algorithms, Web Technologies, Database Systems

**University of Mumbai** GPA: 9.13 / 10

**Jul 2020 - May 2024**

**Bachelor of Engineering in Computer Engineering**

Mumbai, India

Data Structures, Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Natural Language Processing

## EXPERIENCE

**AI Engineer Intern – Search & Recommendation Systems | Tabhi, Austin, TX**

**May 2026 – Aug 2026**

- Architected an end-to-end personalized discovery feed (retrieve, rank, re-rank) for a 450K-item catalog using hybrid Elasticsearch BM25 and vector retrieval with slot-based re-ranking.
- Designed a recency-decayed user preference model to personalize retrieval and support future Two-Tower and SASRec ranking.
- Reduced cold-feed latency from 8s to <500 ms by three-tier caching, cache pre-warming, optimized indexing, and geospatial retrieval.
- Built recommendation analytics using RudderStack to capture user behavior and optimize personalization through engagement metrics.
- Resolved ES-MongoDB data consistency issues and hardened location-aware retrieval, improving feed and recommendation reliability.

**Research Assistant – Real-Time AI Systems | USC, Los Angeles, CA**

**Nov 2025 – May 2026**

- Built a real-time SLAM pipeline combining monocular VO and depth estimation, achieving ~20–25 FPS on noisy indoor data.
- Implemented RANSAC-based feature matching (2500 matches, 2000 inliers), leveraging IMU to improve motion consistency under noise.
- Resolved scale ambiguity using sparse depth and fused VO with GPS in GTSAM, reducing trajectory error from ~95 m to ~1.2 m ATE.
- Debugged sensor noise, calibration drift, and VO failures, improving tracking stability and reducing dropouts by ~30%.

**Machine Learning Intern – Edge Computer Vision | YMT Medical, India**

**Jan 2024 – Aug 2024**

- Redesigned YOLOv8 with a custom P2 head and NWD loss, achieving 0.83 mAP@0.5 on micro-lesions across 2,000+ patient images.
- Built an on-device IQA pipeline using Laplacian blur and HSV glare filtering, pruning 35% of degraded patient uploads.
- Accelerated edge inference by 41% (58ms → 34ms P95) via INT8 quantization and operator fusion with <42MB RAM.
- Mapped 468 MediaPipe facial landmarks to compute deterministic GAGS acne scores offline with an active learning loop.

**Software Engineer Intern – Backend & APIs | Technoriya ERP Solution, India**

**Oct 2023 – Dec 2023**

- Developed Python and Node.js backend services and APIs supporting scalable data pipelines and cloud-based ML workflows.
- Implemented secure authentication and access protocols via Firebase Auth and Firestore rules to ensure data integrity.
- Optimized RESTful APIs and Firestore queries to achieve 450 ms response times under 1,000 concurrent requests for ML integration.
- Streamlined deployment workflows using Vercel and GitHub Actions to improve delivery consistency and reliability.

## PROJECTS

**Eval-Gated Skill Development for a Travel Agent | LangGraph, FastAPI, Streamlit, Langfuse**

**May 2026 – June 2026**

- Built an agent-skills platform: versioned 17-skill registry with release CLI, plus a LangGraph travel agent evaluated on a 466-task bank.
- Shipped A/B evals with 3-tier CI gates scoring skills with vs. without injection (+17 pts top skill, ~\$0.03/PR).
- Added a Thompson-sampling optimizer lifting scores 0.33 to 0.60; 13.9K traces retired a router misrouting 35% of tasks.

**Godot MCP: AI-Native Game Development Interface | TypeScript, Node.js, MCP - [Link](#)**

**April 2026 – Present**

- Built an MCP server enabling LLM agents (Claude, Cursor) to control Godot projects via natural language; used by 10+ active users.
- Developed 30+ tools for scene, script, execution, and asset management with secure path validation and structured routing.
- Implemented CI/CD, auto-config detection, and multi-client integration for end-to-end 2D/3D workflow automation.

**Open Source Contribution – Defillama SDK (RPC Reliability & Failover) | TypeScript, Node.js, Web3 - [Link](#)**

**April 2026**

- Fixed RPC override bug (Fixes #162) by redesigning Chain RPC logic to correctly prioritize user-defined endpoints.
- Implemented runtime RPC quarantine (5-failure threshold, 10 min cooldown), reducing repeated retries and improving request reliability.
- Added 8 unit tests covering override and failure-handling logic; expanded SDK documentation for RPC configuration.

## TECHNICAL SKILLS, PUBLICATIONS, CONFERENCES

- **Programming Languages:** Python, JavaScript, TypeScript
- **AI / ML & Data Science:** Deep Learning, Generative AI, NLP, LLMs (ChatGPT, Claude Code), Transformers, RAG, Agentic AI Systems, Prompt Engineering, Computer Vision, Data Science, PyTorch, LangChain, Pandas, NumPy, LangGraph, MCP, Google ADK, A2A
- **Frameworks & Development:** FastAPI, Flask, Node.js, Express.js, React.js, Next.js, OAuth, HTML/CSS
- **Databases & Cloud:** MongoDB, SQL, PostgreSQL, Supabase, Redis, AWS, GCP, Azure, Docker, Kubernetes, Git, CI/CD, Gitlab, Linux
- **Software Engineering:** Object-Oriented Programming (OOP), REST APIs, Agile
- **Publication:** P. Baradkar et al., "Blockchain Based Music Streaming Platform Using NFTs," IEEE ICSCSS 2023. [Cited by 7]
- **Hackathons:** Syrus '23 Winner, Shortlisted for SIH '23 Hackathon, Indehub '25 Hackathon, CodeShashtra '23 & '24, ICSCSS'23 & SWE'25